Predicting NBA Draft Success:

Can Pre-Draft Measurables and College Statistics Predict Early NBA Performance?

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I. Background and Description of the Problem and Data Set, Response and Predictors

Each year, NBA franchises invest substantial resources into evaluating draft prospects. Scouts analyze game film, attend workouts, and review statistical profiles, yet busts remain common and gems slip through every draft. This project asks a fundamental question:

Can pre-draft information alone, combine measurables and college statistics, predict how well a player will perform in their first two NBA seasons?

We restrict our analysis to players who followed the traditional domestic pathway: high school, at least one year of college, then the NBA draft. We chose this restriction because international players compete in leagues of vastly different quality: a 20-point average in Brazil's NBB is not comparable to 20 points per game in the NCAA. Restricting to college-route players ensures every predictor we use is measured on a standardized, comparable scale.

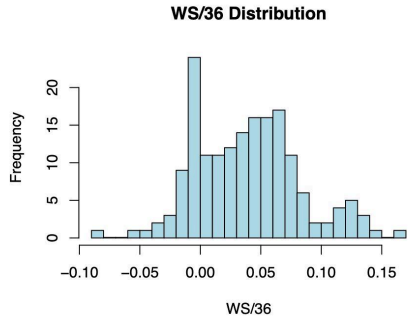
Our sample consists of 132 traditional-route players drawn from the 2022, 2023, and 2024 NBA draft classes, the three most recent classes for which two full seasons of NBA data are available. These draft classes were selected to reflect the modern NBA, characterized by pace-and-space offenses, positionless play, and analytics-driven roster construction. One observation (Shaedon Sharpe, 2022) was removed prior to analysis because he enrolled at Kentucky but never played a game, leaving his college statistics as zeros. Cook's Distance confirmed this observation was exerting disproportionate influence on the model ($D \approx 1.0$, approximately 33 times the threshold of $4/n$), and removing him improved Adjusted R^2 from 0.217 to 0.266.

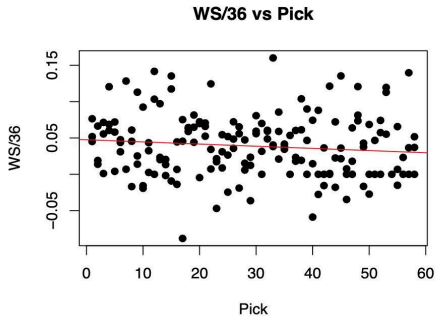
The response variable is Average WS/36, Win Shares per 36 minutes, averaged across a player's first two NBA seasons. Win Shares is a comprehensive single-number metric from Basketball-Reference that estimates how many wins a player contributed to his team. Dividing by minutes played and scaling to 36 minutes normalizes for playing time opportunity, which varies considerably among rookies depending on team context.. Table 1 provides a reference scale for interpreting WS/36 values.

Table 1: WS/36 Reference Scale

WS/36 Range	Performance Level
≤ 0.000	Replacement level or below
0.050 – 0.100	Bench / fringe player
0.100 – 0.150	Solid starter
0.150 – 0.200	All-Star caliber
0.200+	MVP candidate

The histogram of WS/36 shows a roughly symmetric distribution centered near 0.04, ranging from -0.10 to 0.15, with a spike near zero, which reflects players who saw minimal rookie minutes. As for the scatter plots, of WS/36 against draft pick number shows a slightly negative trend with a wide spread at every pick position. Which means that draft pick alone is a weak predictor.





Our **predictors** fall into three categories, all measured before the player's first NBA game. The first category is NBA Combine measurables: wingspan-to-height ratio (a scale-invariant measure of physical length), maximum vertical leap in inches, body fat percentage, and three-quarter court sprint time. The second category is college career averages: True Shooting percentage (TS%), Assist-to-Turnover ratio, Rebounds per 36 minutes (REB36), and Steals per 36 minutes (STL36). The third category consists of contextual variables: age at draft, position (Guard, Wing, or Big, with Big as the reference category to avoid the dummy variable trap), and a binary indicator for whether the player attended a Top 25 nationally-ranked high school program. All data were collected from Basketball-Reference.com and sports-reference.com/cbb.

II: Full Regression Model

We fit the following full multiple linear regression model:

$$WS/36 = \beta_0 + \beta_1(Age) + \beta_2(MaxVert) + \beta_3(WingRatio) + \beta_4(Guard) + \beta_5(Wing) + \beta_6(TS\%) + \beta_7(AST/TOV) + \beta_8(REB36) + \beta_9(STL36) + \beta_{10}(HS) + \varepsilon$$

The full model produced an R^2 of 0.322 and an Adjusted R^2 of 0.266, indicating that the predictors together explain roughly one-third of the variation in WS/36. The overall F-test is highly significant ($F = 5.742$ on 10 and 121 df, $p = 5.06 \times 10^{-7}$), confirming that the model as a whole does substantially better than simply predicting the sample mean.

At the individual predictor level, REB36 was the dominant predictor ($\beta = 0.0096$, $p = 8.86 \times 10^{-6}$), indicating that each additional rebound per 36 minutes in college is associated with approximately 0.010 additional WS/36, holding all else constant. MaxVert was also significant ($\beta = 0.0168$, $p = 0.034$), suggesting players who recorded a vertical leap at or above the 36-inch threshold averaged roughly 0.017 higher WS/36 than those below. TS% was marginally significant ($p = 0.055$). All other predictors, WingRatio, Guard, Wing, Age, STL36, AST/TOV, and HS, did not individually clear the $\alpha = 0.05$ threshold in the full model.

The signs on all coefficients are interpretable. REB36, MaxVert, TS%, AST/TOV, STL36, HS, and Age carry positive coefficients, consistent with the expectation that better rebounding, athleticism, shooting efficiency, ball control, defensive activity, elite high school prestige, and physical maturity should all contribute positively to early NBA production. WingRatio carries a negative coefficient (-0.218) which may appear counterintuitive, as greater wingspan is generally considered an advantage. However, this coefficient is not significant (p = 0.104) and likely reflects multicollinearity with position with bigs. The model output is attached below:

Table 2: Full Regression Output:

```
lm(formula = WS36 ~ Age + TS + ASTTOV + REB36 + STL36 + HS +
    WingRatio + MaxVert + Guard + Wing, data = draft_college)

Residuals:
      Min        1Q    Median        3Q       Max
-0.090658 -0.021744  0.001564  0.023648  0.088982

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.0239713  0.1567731   0.153   0.8787
Age          0.0028375  0.0030732   0.923   0.3577
TS           0.0014797  0.0007636   1.938   0.0550 .
ASTTOV       0.0115443  0.0071616   1.612   0.1096
REB36        0.0095521  0.0020582   4.641 8.86e-06 ***
STL36        0.0053819  0.0072638   0.741   0.4602
HS           0.0139617  0.0086604   1.612   0.1095
WingRatio   -0.2184546  0.1336389  -1.635   0.1047
MaxVert      0.0167907  0.0078392   2.142   0.0342 *
Guard        0.0021406  0.0129649   0.165   0.8691
Wing         0.0096356  0.0103965   0.927   0.3559
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.03741 on 121 degrees of freedom
Multiple R-squared:  0.3218,    Adjusted R-squared:  0.2658
F-statistic: 5.742 on 10 and 121 DF,  p-value: 5.064e-07
```

IIc: Full Model Diagnostics:

(a) Constant Variance. The residuals vs. fitted values plot shows approximately random scatter around zero with no clear fan shape or systematic pattern, suggesting the homoscedasticity assumption is reasonably satisfied. The R-Student residuals plot is consistent with this assessment, with the large majority of values falling comfortably within ±2.

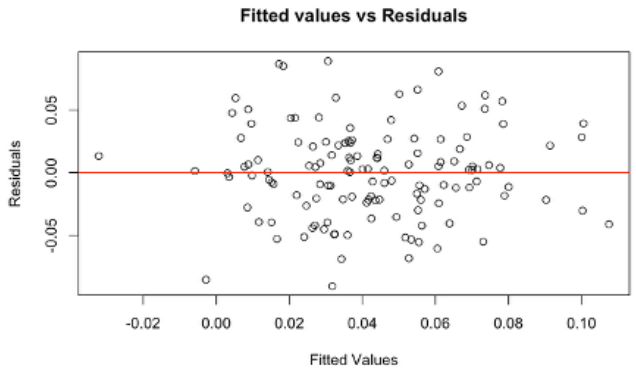
(b) Normality. The Q-Q plot of residuals follows the reference diagonal closely in the central region, with minor departures in both tails indicating slight heavy-tailedness. The Shapiro-Wilk test returned p = 0.59, well above any conventional significance level, so we formally fail to reject normality. The normality assumption is satisfied.

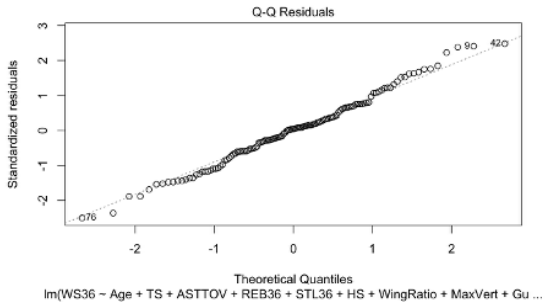
(c) Leverage. Using the threshold 2(k+1)/n = 22/132 = 0.166, four observations were flagged as having high leverage: observations 14, 25, 78, and 90. These players exhibit unusual predictor combinations, for instance, observation 90 had an exceptionally high TS% of 69.9% with a below-average REB36 of 5.1. High leverage alone does not indicate a harmful observation; it indicates an unusual position in predictor space.

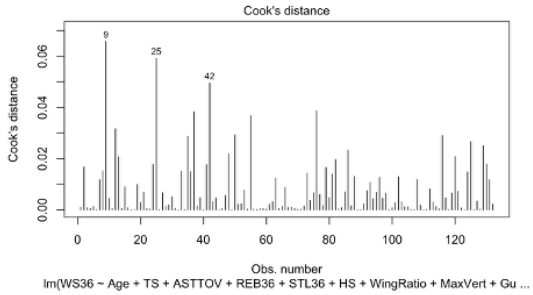
(d) Outliers in Y. Using Bonferroni-corrected jackknife residuals at α = 0.05, no observations were formally identified as outliers in the response. Without the Bonferroni correction, 7 observations were flagged, but this is precisely what we would expect from 132 simultaneous tests under the null (132 × 0.05 ≈ 6.6), confirming these are false positives from multiple testing rather than true outliers.

(e) Influential Points. Prior to removing Shaedon Sharpe, Cook's Distance for observation 7 was approximately 1., roughly 33 times the threshold of 4/n ≈ 0.030. This extreme influence was attributable

entirely to missing college data coded as zeros, distorting every coefficient. After removing Sharpe and refitting, no remaining observation exceeded 0.10 in Cook's Distance. The model with 132 observations (Sharpe removed) was used for all subsequent analyses. **Helpful plots can be seen below:**



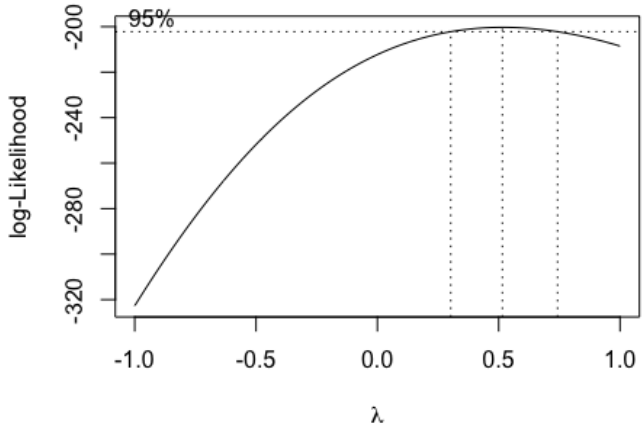




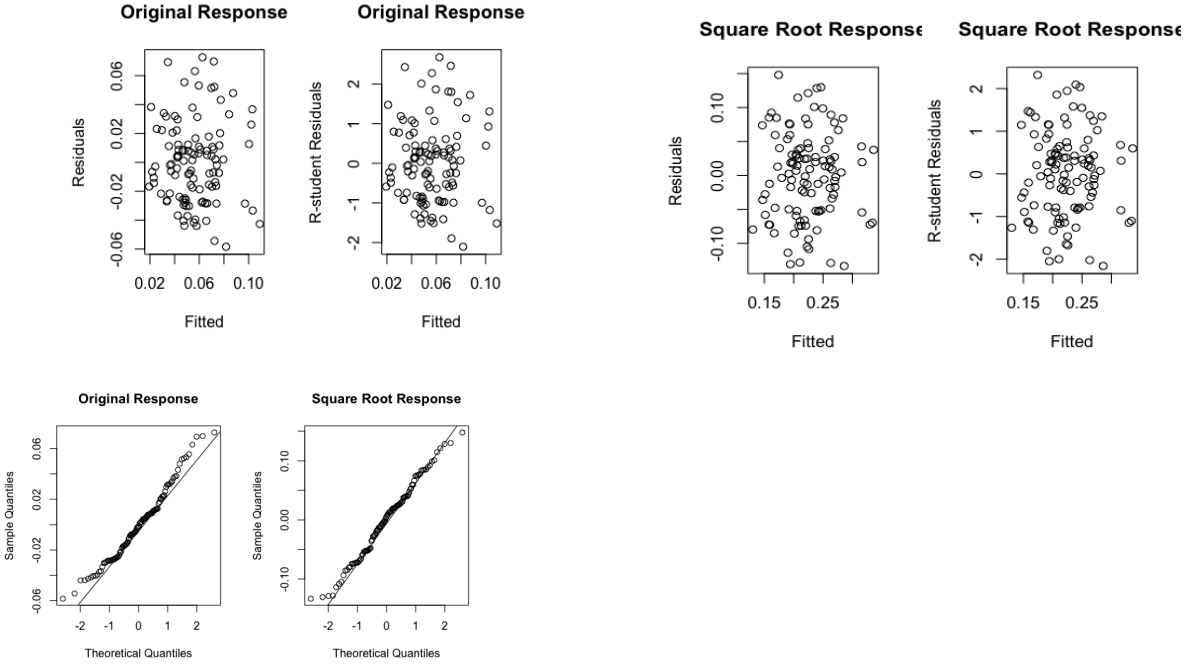
IId: Potential Transformations to Correct Model Inadequacies

Box-Cox Transformation. Although the Shapiro-Wilk test on the full sample passed comfortably ($p = 0.59$), the positive-response subset of 108 players, the 24 players with $WS/36 \leq 0$ were excluded since Box-Cox requires strictly positive responses, returned $p = 0.050$, just at the boundary of concern. We ran a Box-Cox analysis using the MASS package. The profile log-likelihood was maximized near $\lambda = 0.55$, with a 95% confidence interval of approximately (0.35, 0.80). Since $\lambda = 0.5$ (square root) falls within this interval and is interpretable, we refit the model on the 108-player subset using $\sqrt{(WS/36)}$ as the response.

The transformation improved Shapiro-Wilk from $p = 0.050$ to $p = 0.41$, resolving the borderline normality concern. Constant variance remained reasonable before and after the transformation. R^2 was essentially unchanged (0.310 vs. 0.313), and the same predictors remained significant: REB36 ($p = 6.31 \times 10^{-5}$) and MaxVert ($p = 0.014$). TS%, which was already marginal before transformation, weakened further to $p = 0.131$. The transformation did not change the substantive findings.



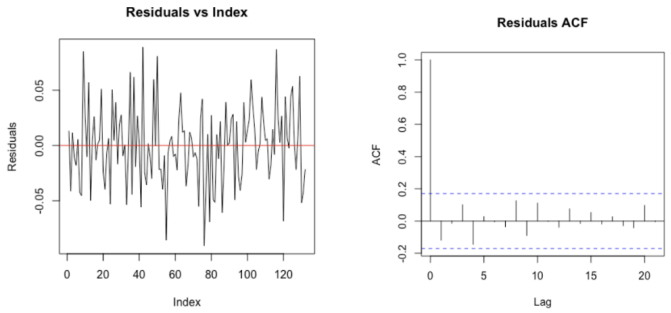
Residual plots before and after the square root transformation



Weighted Least Squares and Serial Correlation:

Weighted Least Squares: Since REB36 was the dominant predictor, we tested whether error variance depended linearly on it following the procedure in Topic 15. Players were grouped into five REB36 bins and group variances were regressed on group means. The slope was -0.0053 ($p = 0.290$), not significant. OLS and WLS coefficients were identical to six decimal places and the weighted residual plot showed random scatter with constant spread. OLS remains appropriate.

Serial Correlation: Observations were sorted by draft year and then pick number. The residual vs. index plot showed random oscillation around zero with no drift, trend, or clustering. The ACF plot confirmed this: all lags beyond lag 0 fell within the 95% confidence bands with no AR(1) decay pattern. The Durbin-Watson test (DW = 2.234, p = 0.914) formally failed to reject the null of no autocorrelation. As a further check, a GLS model with AR(1) correlation structure was fit to the sorted data. The estimated autocorrelation parameter was Phi = -0.150 with 95% CI (-0.330, 0.040), which contains zero, confirming no significant serial dependence. OLS assumptions hold.



IId continued: Multicollinearity

In the full model from Section 2, we saw a highly significant overall F-statistic of 5.742 and a p-value of 5.06e-07, but only REB/36 and MaxVert had p-values less than 0.05 individually at 5% significance, with TS near significance at a p-value of 0.055. This isn't the collinearity warning we learned in lecture of a significant F-statistic with no significant t-statistics, but the gap between the strong overall fit and the weakness of most individual predictors still made it worth checking.

The pairwise correlation matrix did not show anything alarming. The biggest correlations were Age and HS at −0.591, REB/36 and Guard at −0.556, and Guard and Wing at −0.544. The Age and HS one makes sense because top-25 high school players tend to enter the draft younger, the REB/36 and Guard one makes sense because guards do not rebound as much as bigs (the reference group), and Guard and Wing makes sense because those two are mutually exclusive position dummies. Also, nothing came close to the 0.9 range that would actually be a red flag.

The eigenvalues of X′X ran from about 5.12×10^5 down to 0.346, which gave a largest condition index of 1216. That looks alarming at first, but the inflation is mostly due to scaling. Age is in years, TS is in percentage points, WingRatio sits near 1, and Guard, Wing, and HS are 0/1 dummies. Eigenvalues are not scale-invariant, so the condition indices blow up from the units rather than from any real near-linear dependence between predictors. For this reason we decided to trust the pairwise correlations and the VIFs more.

Looking at the VIFs, they were all below 5 which is well under the usual cutoff of 10. The biggest was Guard at 3.60, followed by REB/36 at 2.38 and Wing at 2.34, which lines up with the pairwise correlations from above so these are mild and consistent with the pairwise results. We believe that multicollinearity is not the reason most of the predictors fail to clear significance individually. The unexplained variation in WS/36 ($R^2 = 0.3218$) reflects the genuine difficulty of predicting player performance from pre-draft data, not predictors stepping on one another.

Table 3: VIF and Multicollinearity Matrix

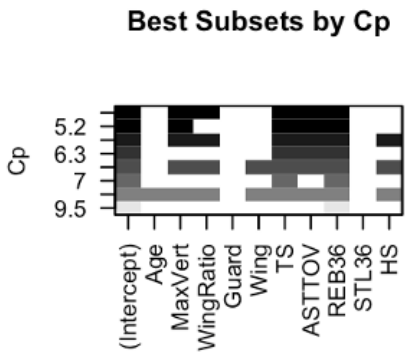
```
             Age     TS ASTTOV  REB36  STL36     HS WingRatio MaxVert  Guard   Wing   WS36
Age        1.000  0.355  0.281  0.170  0.013 -0.591    -0.018  -0.160 -0.104  0.020  0.168
TS         0.355  1.000  0.058  0.364 -0.033 -0.290     0.009  -0.111 -0.250 -0.090  0.323
ASTTOV     0.281  0.058  1.000 -0.287  0.325 -0.221    -0.041  -0.055  0.410 -0.166  0.006
REB36      0.170  0.364 -0.287  1.000 -0.130 -0.166     0.076  -0.136 -0.556 -0.104  0.479
STL36      0.013 -0.033  0.325 -0.130  1.000 -0.074     0.066   0.133  0.250 -0.025  0.042
HS        -0.591 -0.290 -0.221 -0.166 -0.074  1.000     0.024   0.013  0.150 -0.124 -0.085
WingRatio -0.018  0.009 -0.041  0.076  0.066  0.024     1.000   0.239 -0.101  0.027 -0.044
MaxVert   -0.160 -0.111 -0.055 -0.136  0.133  0.013     0.239   1.000  0.052  0.042  0.043
Guard     -0.104 -0.250  0.410 -0.556  0.250  0.150    -0.101   0.052  1.000 -0.544 -0.262
Wing       0.020 -0.090 -0.166 -0.104 -0.025 -0.124     0.027   0.042 -0.544  1.000 -0.018
WS36       0.168  0.323  0.006  0.479  0.042 -0.085    -0.044   0.043 -0.262 -0.018  1.000
 [1] 5.115650e+05 7.101868e+02 3.923932e+02 5.453558e+01 3.801938e+01 3.327843e+01 2.451788e+01 1.841861e+01 6.304241e+00 3.459255e-01
 [1]    1.00000   26.83886   36.10686   96.85241  115.99731  123.98495  144.44716  166.65639  284.86167 1216.07153
      Age        TS    ASTTOV     REB36     STL36        HS WingRatio   MaxVert     Guard      Wing
 1.762294  1.357106  1.553882  2.379634  1.216212  1.664840  1.089845  1.149812  3.600008  2.337792
```

IIe: Model Selection

Exhaustive Search: Using regsubsets() from the leaps package, we evaluated all $2^{10} = 1{,}024$ possible subsets of our ten predictors. Four criteria were computed for each model size. Adjusted R^2 plateaued around 6-8 parameters, with diminishing returns beyond p = 6. Mallow's Cp was minimized at p = 5 (Cp ≈ 5.0, just below the diagonal indicating near-unbiased prediction relative to the full model). AIC was clearly minimized at p = 6. BIC, which carries a heavier penalty towards the sample size and the number of predictors, favored the smallest models (p = 2-3). AIC and Cp reached strong consensus on a 5-predictor model, which can be seen below in table 4.

Table 4: Model Selection Results by Criterion

Criterion	Optimal Size	Selected Predictors
BIC	2 params	REB36 only
AIC	6 params (5 predictors)	MaxVert, WingRatio, TS%, AST/TOV, REB36
Cp	5 params (4 predictors)	MaxVert, WingRatio, TS%, AST/TOV, REB36
Adj. R^2	8 params (7 predictors)	MaxVert, WingRatio, Wing, TS%, AST/TOV, REB36, HS



Using backward elimination, we started off with the full model (adjusted R-squared = 0.2658, F-test p-value = 5.06e-07 – significant) and removed them at a time in order of highest p-value: Guard, STL36, Age, Wing, HS, WingRatio, MaxVert, then ASTTOV. The final model with all variables significant at alpha = 0.05 contains REB36 and TS to explain WS36, REB36 remaining strong throughout eliminations. The final two-predictor model has an adjusted R-squared of 0.2432, compared to 0.2658 for the full model, meaning that we lost relatively little explanatory power while dropping eight predictors. This suggests that college rebounding rate (REB36) and true shooting percentage (TS) together capture the most meaningful pre-draft signal for predicting early NBA production among college-route players.

Using forward selection, we added predictors one at a time by the smallest remaining p-value. REB36 entered first as the strongest single predictor at p-value = 6.255e-09, followed by TS at a p-value = 0.038. After adding both these predictors, no remaining predictors were below the alpha = 0.05, so selection stopped. The final model of REB36 and TS matches the final product of backward elimination. This agreement gives confidence that these two variables are the most meaningful predictors of rookie WS36 among college draft picks in this dataset.

IIf: Diagnostics of the Reduced Model

(a) Constant Variance (Homoscedasticity)
The residuals vs. fitted values plot shows that the spread of residuals is constant across fitted values. The R-student residual plot similarly shows no clear pattern.
Conclusion: There is no evidence of heteroscedasticity, so the constant variance assumption is reasonable.

(b) Normality of Errors
The QQ-plot of residuals shows that the points slightly deviate the reference line, particularly in the tails. The Shapiro-Wilk test yields a p-value of 0.439.
Conclusion: Since the p-value is greater than 0.05, we fail to reject normality. Overall, the normality assumption is reasonable.

(c) Large Leverage Points
Using the cutoff 2 (p/n), the following observations were identified as having high leverage: 2, 12, 23, 25, 65, 90, 94, 112, 127 and 131.
Conclusion: These points do not appear to exert unusual influence on the model due to their extreme predictor values.

(d) Outliers in y
Using the R-student residual criterion, the following observations were identified as outliers: none (but 10 are flagged via r_student graph).
Conclusion: These observations may be unusual in terms of their response values relative to the model.

(e) Influential Points
Using Cook's Distance with cutoff 1, the following observations were identified as influential: none.
Conclusion: These points do not have a strong impact on the fitted model and should not be of concern.

III: Conclusions and Interpretations

The primary finding of this analysis is that college rebounding rate (REB36) is by far the most robust predictor of early NBA success among traditional-route players. REB36 was significant in the full model ($p = 8.86 \times 10^{-6}$), remained significant through every step of backward elimination, was the first predictor selected in forward selection ($p = 6.26 \times 10^{-9}$), and appeared in every model selected by every criterion at every model size in the exhaustive search. A player who grabbed one additional rebound per 36 minutes in college is predicted to add approximately 0.009-0.010 WS/36 in their first two NBA seasons, holding all else constant.

Maximum vertical leap (MaxVert) emerged as a significant predictor in the full model ($p = 0.034$) and was retained by the AIC/Cp exhaustive search. Its significance was obscured in the initial full model fit that included Shaedon Sharpe, whose zeros artificially suppressed this coefficient, a concrete illustration of how a single influential observation with data quality problems can distort inference. After removing Sharpe, MaxVert's coefficient became significant and stable.

True Shooting percentage (TS%) presented an interesting pattern. It was marginally significant in the full model ($p = 0.055$), became the second predictor selected by both backward and forward selection ($p = 0.038$ at the step of entry), and was retained by both the AIC/Cp and backward/forward reduced models. While it does not consistently clear $\alpha = 0.05$ across all model specifications, the convergence of multiple selection methods on TS% as the second most informative predictor after REB36 suggests it carries a real, if modest, predictive signal.

Position dummies (Guard and Wing) were dropped early in every selection procedure and carried near-zero, non-significant coefficients in the full model. This suggests that once college statistical production and physical measurables are controlled for, positional labels adds no independent information about WS/36. The statistical profile captures what position would otherwise proxy for.

The AIC/Cp five-predictor model, WS/36 ~ MaxVert + WingRatio + TS% + AST/TOV + REB36, achieved Adjusted $R^2 = 0.272$, $F(5, 126) = 10.77$, $p = 1.23 \times 10^{-8}$, RSE = 0.0373. This represents a slight improvement over the full model in both Adjusted R^2 and RSE despite using only five predictors. The backward/forward two-predictor model, WS/36 ~ REB36 + TS%, achieved Adjusted $R^2 = 0.243$, retaining most explanatory power with maximum parsimony.

IV: Conclusions

This analysis demonstrates that pre-draft information has meaningful but limited predictive power for early NBA success among college-route players. Our best models explain approximately 24-27% of the variation in WS/36, a non-trivial amount given the inherent unpredictability of athletic development, but also a reminder that the majority of variation in rookie performance cannot be captured by any pre-draft observable.

College rebounding rate is the single most actionable finding. REB36 predicts early NBA production more reliably than any physical measurable, any other college statistic, or any contextual variable in our dataset. This is consistent with the broader analytics literature suggesting that rebounding,particularly among guards and wings who rebound at above-positional-average rates, is one of the most translatable college skills to the professional level. Scouts and front offices should weigh college rebounding heavily when evaluating prospects.

The significance of MaxVert suggests that raw athleticism, measured precisely and objectively at the combine, does carry predictive information beyond what college statistics capture. However, its effect size is modest, and it was only detectable after correcting a data quality issue, underscoring that clean, reliable data is a prerequisite for valid inference.

Several limitations should be noted. First, restricting to college-route players excludes international prospects, who represent a growing share of top NBA talent. Our findings do not generalize to players who bypassed college entirely. Second, our sample of three draft classes (n = 132) is relatively small for the number of predictors considered. Third, WS/36 is itself an imperfect metric, it relies on team defensive ratings and can be influenced by circumstances beyond a player's control. Future work could extend this framework to international players using league adjustment factors, incorporate more draft classes for greater power, or explore non-linear models to capture more complex predictor relationships.

See appendix below:

Appendix: All Corresponding R Code:

Task 1: Data Set-Up Code

```r
# Step 1
library(dplyr)

draft <- read.csv("nba_data1.csv")

# Rename columns for easy naming
names(draft)[names(draft) == "Average.WS.36"] <- "WS36"
names(draft)[names(draft) == "Draft.Pick"] <- "Pk"
names(draft)[names(draft) == "Traditional.route..1.if.yes."] <- "trad"
names(draft)[names(draft) == "Draft.Class"] <- "draft_yr"
names(draft)[names(draft) == "Age.at.Draft"] <- "Age"
names(draft)[names(draft) == "HS..Top.25.School..1.yes..0.otherwise."] <- "HS"
names(draft)[names(draft) == "Wingspan..in..at.DRAFT"] <- "Wingspan"
names(draft)[names(draft) == "Height..in..AT.DRAFT"] <- "Height"
names(draft)[names(draft) == "Max.Vertical..1.if.Player.had.Vertical...36.in..0.otherwise"] <-
names(draft)[names(draft) == "Guard..1."] <- "Guard"
names(draft)[names(draft) == "Wing.0."] <- "Wing"
names(draft)[names(draft) == "TS."] <- "TS"
names(draft)[names(draft) == "AST.TOV"] <- "ASTTOV"
names(draft)[names(draft) == "REB.36"] <- "REB36"
names(draft)[names(draft) == "STL.36"] <- "STL36"


# Draft indicator: 1 = college, 0 = international
draft$intl <- ifelse(draft$trad == 0, 0, 1)
draft$WingRatio <- draft$Wingspan/draft$Height
draft_college <- draft %>%
    filter(trad == 1)
```

```r
# Scatter Plot
plot(WS36 ~ Pk, data = draft, main = "WS/36 vs Pick", xlab = "Pick", ylab = "WS/36", pch = 19)
abline(lm(WS36 ~ Pk, data = draft), col = "red")
```

Task 2 Corresponding Code:

```r
# Step 2:Fit the full multiple linear regression model

draft_model <- lm(WS36 ~ Age + TS + ASTTOV + REB36 + STL36 + HS +
                    WingRatio + MaxVert + Guard + Wing,
                  data = draft_college)
summary(draft_model)
```

Task 3 Corresponding Code:

```r
# Step 3: Checking Model Diagnostics!

#part a) check the constant variance assummption
plot(fitted(draft_model), resid(draft_model), xlab = 'Fitted Values',
     ylab = 'Residuals', main = 'Fitted values vs Residuals')
abline(h = 0, col = "red")
#Interpretation:
# Values are roughly scattered around zero, with a random-looking scatter, no pattern
# However, there are a few extereme negative and positive valuesm likely high usage or low usage players
# We would say it's not perfectly satisfied, but reasonably satisfied

#a2) check the R-Student residuals
plot(fitted(draft_model), rstudent(draft_model), xlab = 'Fitted Values',
     ylab = 'R-Student Residuals', main = 'Fitted Values vs R-Student')
abline(h = 0, col = 'red')
# the bounds of the residuals fall reasonably within two horizontal bands
# general random scatter
# we see one exit this band who also seems to have the most unique fitted value
# they might be a high leverage or an influential point on the model!

#part b) checking normality
plot(draft_model, which = 2)
#Interpretation: we see a big chunk of the middle data is approximately normal
# however, what we have as seen with the slight concavity is heavier tails
# this implies we have players who were exceedingly over or under performing on both ends
# essentially, their pre-draft measurements showed them to be average, and they outperformed or
# underperformed their theoretical statistics

#part b2: shapiro-wilk test
shapiro.test(resid(draft_model))
#output: W = 0.99, p-value = 0.59, no statistical evidence that our residuals depart from normality
```

‘

```r
#part c) Check for large leverage points
X <- model.matrix(draft_model)
sigma_sq <- sigma(draft_model)^2
H <- X%*%vcov(draft_model)%*%t(X) / sigma_sq
leverage<- diag(H)
leverage
# check if points have leverage > 2 * (k+1)/n
# here, k+1 = 11, n = 132. so 2 * (11) / 132 = 0.166
which(leverage > 0.165) # the result is obs 14, obs 90


#obs 14 has leverage 0.206, for player Tari Eason. He had exceptional rebounding and Stl Statistics

# obs 90 has leverage 0.1963 for player Reed Shappard, who also had solid Rebounding and Stl Stats

#part d: checking for outlier points in y
# we use the jackknife residuals, and then we compare using the bonferonni correction
# which will be alpha/2*n, n-p, p here = 11, n = 133
jackknife = rstudent(draft_model)
qt(1-(0.05/(2*132)),132-11)
which(abs(jackknife) >= qt(1-(0.05/(2*132)),132-11))
# confirming normality of residulas, no outliers in y were found using the correction
#w/o bonferonni
which(abs(jackknife) >= qt(1-(0.05/(2)),132-11))
# obs 7, 10, 13, 43, 51, 56, 77, 117. given we have alpha = 0.05 and n = 133,
# we expect 6 false positives to begin with.

#part e) influential points
cook = cooks.distance(draft_model)
which(cook > 0.5)
```

```r
plot(draft_model, which = 4)
# as you can see not a single other point goes even above 0.2!!!

draft_college[90,]
```

Task 4 Corresponding Code:

```r
# Step 4

library(MASS)
library(dplyr)

draft_pos <- draft_college %>% filter(WS36 > 0)

g <- lm(WS36 ~ Age + TS + ASTTOV + REB36 + STL36 + HS +
          WingRatio + MaxVert + Guard + Wing, data = draft_pos)
summary(g)

boxcox(g, plotit = T)
boxcox(g, plotit = T, lambda = seq(-1, 1, by = 0.05))

# Square root transformation (lambda = 0.5, inside the 95% CI)
g2 <- lm(sqrt(WS36) ~ Age + TS + ASTTOV + REB36 + STL36 + HS +
           WingRatio + MaxVert + Guard + Wing, data = draft_pos)
summary(g2)

# Compare original vs transformed diagnostics (matches Galapagos slide 19)
par(mfrow = c(1,2))
plot(fitted(g),  residuals(g),  xlab = "Fitted", ylab = "Residuals",
     main = "Original Response")
plot(fitted(g),  rstudent(g),   xlab = "Fitted", ylab = "R-student Residuals",
     main = "Original Response")

par(mfrow = c(1,2))
plot(fitted(g2), residuals(g2), xlab = "Fitted", ylab = "Residuals",
     main = "Square Root Response")
plot(fitted(g2), rstudent(g2),  xlab = "Fitted", ylab = "R-student Residuals",
     main = "Square Root Response")
par(mfrow = c(1,1))

# Normality check before vs after
shapiro.test(residuals(g))
shapiro.test(residuals(g2))

par(mfrow = c(1,2))
qqnorm(residuals(g),  main = "Original Response"); qqline(residuals(g))
qqnorm(residuals(g2), main = "Square Root Response"); qqline(residuals(g2))
par(mfrow = c(1,1))
```

Task 5 and 6 Corresponding Code:

```r
# Step 5
library(MASS)
library(dplyr)

# Sort by draft year then pick number
draft_sorted <- draft_college[order(draft_college$draft_yr, draft_college$Pk), ]

# Refit OLS on sorted data
ols_sorted <- lm(WS36 ~ Age + MaxVert + WingRatio + Guard + Wing + TS + ASTTOV + REB36 + STL36 + HS, data =
draft_sorted)

# Check for serial correlation
par(mfrow = c(1, 2))
plot(resid(ols_sorted), type = "l", main = "Residuals vs Index",
     xlab = "Index", ylab = "Residuals")
abline(h = 0, col = "red")
acf(resid(ols_sorted), main = "Residuals ACF")
```

```r
# Formal test
install.packages("lmtest")
library(lmtest)
dwtest(ols_sorted)
```

```r
# GLS correction
install.packages("nlme")
library(nlme)

gls_model <- gls(WS36 ~ Age + MaxVert + WingRatio + Guard + Wing + TS + ASTTOV + REB36 + STL36 + HS, data =
draft_sorted, correlation = corAR1(form = ~1), method = "ML")

summary(gls_model)
intervals(gls_model)
```

```r
# Step 6: Weighted Least Squares

# Group by REB36 bins to estimate variance structure
draft_college$reb_grp <- cut(draft_college$REB36, breaks = 5)

grp_stats <- tapply(draft_college$WS36, draft_college$reb_grp, function(x)
  c(mean = mean(x), var = var(x)))

grp_mean <- sapply(grp_stats, function(x) x["mean"])
grp_var  <- sapply(grp_stats, function(x) x["var"])

# Regress group variances on group means (Topic 15 slide 40)
var_model <- lm(grp_var ~ grp_mean)
summary(var_model)

# Estimate weights for each observation
var_hat <- var_model$coef[1] + var_model$coef[2] * draft_college$REB36
var_hat <- pmax(var_hat, 1e-6)  # avoid zero/negative weights
w <- 1 / var_hat

# Fit WLS model
fit_wls <- lm(WS36 ~ Age + TS + ASTTOV + REB36 + STL36 + HS +
              WingRatio + MaxVert + Guard + Wing,
              data = draft_college, weights = w)
summary(fit_wls)

# Compare OLS vs WLS coefficients
round(cbind(OLS = coef(draft_model), WLS = coef(fit_wls)), 6)

# Plot weighted residuals vs weighted fitted values (Topic 15 slide 44)
plot(sqrt(w) * fitted(fit_wls), sqrt(w) * residuals(fit_wls),
     xlab = expression(w^{1/2} * hat(y)),
     ylab = expression(w^{1/2} * e),
     main = "Weighted Residuals vs Weighted Fitted")
abline(h = 0, lty = 2)
```

Task 7 Corresponding Code:

```r
# Step 7: Assess Multicollinearity (Topic 16)

library(faraway)

# Pairwise correlation matrix of the predictors and response
round(cor(draft_college[, c("Age","TS","ASTTOV","REB36","STL36","HS",
                            "WingRatio","MaxVert","Guard","Wing","WS36")]), 3)

# Eigenvalues of X'_*X_* and condition indices
x <- model.matrix(draft_model)[, -1]
e <- eigen(t(x) %*% x)
e$val
sqrt(e$val[1] / e$val)

# Variance Inflation Factors
vif(x)
```

Task 8 Corresponding Code:

```r
# Task 8: Exhaustive Model Selection
library(leaps)

# Run exhaustive search
b <- regsubsets(WS36 ~ Age + MaxVert + WingRatio +
                  Guard + Wing + TS + ASTTOV +
                  REB36 + STL36 + HS,
                data = draft_college_FILTER,
                nbest = 1,
                method = "exhaustive")

rs <- summary(b)
rs

# Setup
n <- nrow(draft_college)
p <- rowSums(rs$which)

# Compute AIC and BIC manually
aic <- n * log(rs$rss / n) + 2 * p
bic <- n * log(rs$rss / n) + p * log(n)

# Plot all four criteria
par(mfrow = c(2, 3))

# Adjusted R-squared
plot(p, rs$adjr2,
     xlab = "No. of Parameters",
     ylab = "Adjusted R-squared",
     main = "Adjusted R-squared")

# Cp
plot(p, rs$cp,
     xlab = "No. of Parameters",
     ylab = "Cp",
     main = "Mallow's Cp")
abline(0, 1)
```

```
# AIC
plot(p, aic,
     xlab = "No. of Parameters",
     ylab = "AIC",
     main = "AIC")
lines(p, aic)

# BIC
plot(p, bic,
     xlab = "No. of Parameters",
     ylab = "BIC",
     main = "BIC")
lines(p, bic)

# Visual plots
plot(b, scale = "adjr2", main = "Best Subsets by Adj R-squared")
plot(b, scale = "Cp", main = "Best Subsets by Cp")

# Print best model under each criterion
cat("Best model by Adj R-squared - No. of parameters:", p[which.max(rs$adjr2)], "\n")
cat("Best model by Cp - No. of parameters:", p[which.min(rs$cp)], "\n")
cat("Best model by AIC - No. of parameters:", p[which.min(aic)], "\n")
cat("Best model by BIC - No. of parameters:", p[which.min(bic)], "\n")

# Show which variables are in the best BIC model
cat("\nVariables in best BIC model:\n")
print(rs$which[which.min(bic), ])

# Show which variables are in the best AIC model
cat("\nVariables in best AIC model:\n")
print(rs$which[which.min(aic), ])
```

```
# Show which variables are in the best BIC model
cat("\nVariables in best BIC model:\n")
print(rs$which[which.min(bic), ])

# Show which variables are in the best AIC model
cat("\nVariables in best AIC model:\n")
print(rs$which[which.min(aic), ])

# Show which variables are in the best AdjR2 model
cat("\nVariables in best AdjR2 model:\n")
print(rs$which[which.max(rs$adjr2), ])

# Show which variables are in the best Cp model
cat("\nVariables in best Cp model:\n")
print(rs$which[which.min(rs$cp), ])
```

Task 9 and 10 Corresponding Code:

```r
# Step 9
library(dplyr)

#Backward Induction

#Fit the full model
full <- lm(WS36 ~ Age + MaxVert + WingRatio + Guard + Wing + TS + ASTTOV + REB36 + STL36 + HS, data =
draft_college)
summary(full)
```

```r
#Subtract Guard (Highest p-value)
g <- update(full, . ~ . - Guard)
summary(g)
```

```r
#Subtract STL36 (Highest p-value)
g <- update(g, . ~ . - STL36)
summary(g)
```

```r
#Forward Selection

#Intercept-only model
model.current <- lm(WS36 ~ 1, data = draft_college)

#Test all single additions
add1(model.current, ~ Age + MaxVert + Wingspan + Guard + Wing + TS + ASTTOV + REB36 + STL36 + HS, test = "F")
```

```r
#Add the predictor with the SMALLEST p-value (if < 0.05)
model.current <- lm(WS36 ~ REB36, data = draft_college)
add1(model.current, ~ Age + MaxVert + Wingspan + Guard + Wing + TS + ASTTOV + REB36 + STL36 + HS, test = "F")
```

```r
# Step 10
new_draft_model <- lm(WS36 ~ REB36 + TS, data = draft_college)
summary(new_draft_model)
plot(new_draft_model)
n <- nobs(new_draft_model)
p <- length(coef(new_draft_model))
df <- n-p-1

plot(fitted(new_draft_model), rstudent(new_draft_model),
     xlab = "Fitted values", ylab = "R-student residuals",
     main = "R-student Residuals vs Fitted")
abline(h = c(-2, 0, 2), lty = 2, col = "red")

# (b) Normality
shapiro.test(residuals(new_draft_model))

# (c) Large leverage points
which(hatvalues(new_draft_model) > 2 * p / n)

# Plot leverage
plot(hatvalues(new_draft_model), ylab = "Leverage", main = "Leverage Plot")

# Add cutoff line
abline(h = 2 * p / n, col = "red", lty = 2)

# (d) Outliers in y
alpha <- 0.05
alpha_bonf <- alpha / n
t_crit <- qt(1 - alpha_bonf/2, df)

# Identify outliers
which(abs(rstudent(new_draft_model)) > t_crit)

# (e) Influential points
which(cooks.distance(new_draft_model) > 1)
plot(cooks.distance(new_draft_model), type = 'h',
     ylab = "Cook's Distance",
     main = "Cook's Distance Plot")
```